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Towards continual low-light image enhancement through causal inference

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ABSTRACT

Low-light image enhancement focuses on improving the visual perceptual quality and restoring structure details for images captured from poor lightness conditions. The existing approaches often perform well under the identical lightness distribution with the training dataset. We address a continual low-light image enhancement task for the first time to challenge their generalization ability. Specifically, a lightness enhancement network learns a sequence of tasks with different lightness distributions. While the model's generalization performance has been improved, we still identify the presence of catastrophic forgetting in this scenario, which prompts us to further disclose the intrinsic reason from theory. Therefore, we leverage causal inference theory to construct a structural causal model, identifying confounding variables and mitigating their adverse effects through backdoor adjustment, which facilitates the model to learn invariant representation for continual low-light image enhancement. In particular, the Rehearsal-based Invariant Structure Regularizer is employed to enlarge the available value set of the variable for backdoor adjustment. Moreover, a novel Channel Fourier Transform-based Self-Attention module is proposed to enhance the model's generalization ability toward low-frequency information and facilitate the precise estimation of the causal effect. Extensive experiments show the effectiveness in alleviating catastrophic forgetting and superior generalization performance of our method on continuous low-light image enhancement benchmarks.

1. Introduction

Images captured under insufficient lightness conditions exhibit significant quality degradation, leading to not only unpleasant visual results but also reduced performance in many computer vision systems. To remedy this deficiency, low-light image enhancement (LIE) emerges and achieves remarkable success by leveraging the guidance of optical prior knowledge (Guo & Hu, 2023; He et al., 2025) and deep learning methods (Lv et al., 2021), particularly in single-task learning on large-scale datasets, such as LOL-v1 (Wei et al., 2018a) and LOL-v2 (Yang et al., 2021). However, the benchmark methods suffer from limited generalization capability across inconsistent datasets and lightness conditions, and thus remain unsatisfactory for practical application scenarios. In this regard, domain generalization approaches are introduced to improve the LIE model performance across diverse and unseen domains by learning domain-invariant features. A typical and fundamental assumption behind domain generalization is the availability of multiple source domains during training. Therefore, these approaches encounter

challenges in enhancing low-light images while continually adapting to constantly changing lightness conditions.

In this context, Continual Learning (CL), also called lifelong learning or incremental learning, has become a prominent approach to enhance model generalization in continuously changing scenarios, which is dedicated to jointly improving the model plasticity and overcoming catastrophic forgetting (McCloskey & Cohen, 1989). However, previous research on CL has predominantly focused on discriminative tasks, such as classification and recognition, while relatively less attention has been devoted to low-level computer vision tasks, particularly involving LIE. Inspired by the concept of CL, we intuitively explore to achieve a generalized LIE approach under a CL framework, i.e., continual LIE. However, a non-trivial gap between conventional CL tasks and the target task halts the direct adaptation of CL on the LIE within constantly changing lightness conditions, i.e., the semantic information is widely variant among the sequential domains in conventional CL, but in the target task, only the lightness condition-related information remains inconstant, resulting in the relatively limited inconsistency of domains during learning.

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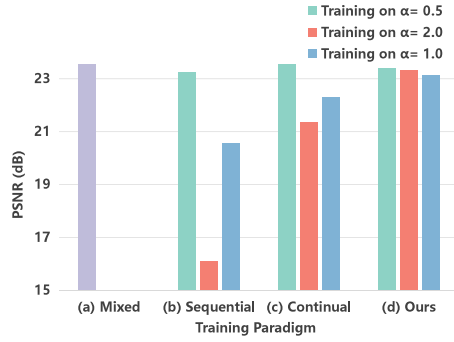


Fig. 1. Comparisons between different training paradigms based on the shared backbone network, i.e., FocalNet (Cui et al., 2023a). α denotes the lightness adjustment coefficient. (a) Mixing images with different lightness conditions into a single dataset. (b) Sequential training on three sets with different lightness conditions. (c) Continuous training using a rehearsal-based CL method. (d) Continuous training using our proposed method. For (b), (c), and (d), α is valued by the ordered set {0.5, 2.0, 1.0} during training, e.g., the model is primarily trained on $\alpha = 0.5$, and the model is subsequently trained on $\alpha = 2.0$ without the parameter initialization. All evaluations are on $\alpha = 0.5$.

This naturally raises the following pivotal questions:

- *Can CL well adapt to the constantly changing lightness conditions?*
- *Does there exist further deficiency hinder the generalization of continual LIE?*

Guided by the focal exploratory questions, we propose to determine the rationality and effectiveness of introducing the CL scheme to achieve low-light enhancement within constantly changing lightness conditions. Thus, a motivational experiment, encompassing various benchmark training paradigms, is conducted. In Fig. 1, α , as the lightness adjustment coefficient (Wang et al., 2023), is introduced to generate brighter ($\alpha = 2.0$) and darker ($\alpha = 0.5$) lightness conditions on LOL dataset (Wei et al., 2018a). Specifically, for (a), we directly train the model on a *mixed* dataset with three different lightness conditions. For (b), we train the model by adopting a sequential scheme without a well-designed learning paradigm. For (c), the model incorporates a benchmark rehearsal-based CL method, i.e., ER (Chaudhry et al., 2019). Evaluation results in (b) and (c) reveal that the final task training on $\alpha = 1.0$ enhances the model's performance on $\alpha = 0.5$ compared to the second task training on $\alpha = 2.0$. This observation suggests that, regardless of whether a CL strategy is employed, continual learning improves the model's generalization capability towards lightness variations. However, there is a significant generalization performance degeneration due to the catastrophic forgetting as we can see in (b) and (c). Thus, continual learning can enhance the model's adaptation to evolving lighting conditions, yet catastrophic forgetting poses a significant challenge that limits its generalization capabilities.

To explore the intrinsic reason behind the challenge stemming from the catastrophic forgetting in continual LIE, we revisit the learning paradigm of the rehearsal-based methods in continual LIE from a causal perspective. Accordingly, we decompose the low-light image in the frequency domain by leveraging the widely-studied Fourier Transform (Yu et al., 2022) to determine the candidate variables, thereby building a Structural Causal Model (SCM) (Pearl et al., 2016). From the SCM, we identify the confounding variable that hinders the improvement of generalization in continual LIE, and further propose a backdoor adjustment-based deconfounding approach to mitigate the negative effect of the confounding variable. To underpin the sufficient deconfounding, the memory buffer from the rehearsal-based CL framework is introduced to enlarge the available value set for adjustment variables, which is implemented by a Rehearsal-based Invariant Structure Regularizer (RISR) in practice. On top of this, holding a fundamental proposition: the lightness information of images is often identified as low-frequency components

in the frequency domain (Cui et al., 2024), we propose a novel Channel Fourier Transform-based Self-Attention module (CFTSA) to enhance the model's generalization ability toward low-frequency information, thereby achieving a precise estimation of deconfounded causal effect. Concretely, our method employs the Global Weighted Average Pooling (GWAP) mechanism, exhibiting the homogeneous characteristics with low-pass filtering (Cui et al., 2023b), to implement the low-frequency information-oriented self-attention module. Consequently, we incorporate the proposed components to derive a plug-and-play approach, namely *Causal Continual Low-light Image Enhancement* (CCLIE), empowering the benchmark methods to improve the generalization to diverse lightness conditions. Benefiting from the plug-and-play characteristic, CCLIE can be feasibly integrated into various network architectures and improve the model generalization. Our crucial contributions can be summarized as follows:

- We disclose an innate defect of benchmark LIE methods: the limited generalization capability across inconsistent datasets and lightness conditions exacerbates the model performance in practical application scenarios.
- We present the SCM-based theoretical analysis to model the learning paradigm of continual LIE, and further provide a causality-guided methodology to address the inherent confounding effects.
- To practically implement the deconfounded causal inference, a novel causal continual low-light image enhancement approach is proposed to explore the causation between low-light images and lightness-enhanced images.
- Extensive experiments demonstrate that the proposed method can alleviate catastrophic forgetting and effectively improve the performance of the state-of-the-art methods on LIE benchmarks.

The structure of this paper is organized as follows. Section 2 reviews related studies on the current generalized low-light image enhancement and continual learning. In Section 3, we present the formulation of the continual low-light image enhancement and the corresponding SCM diagram. Section 4 introduces the proposed method, including the Rehearsal-based Invariant Structure Regularizer and Channel Fourier Transform-based Self-Attention. In Section 5, we provide details of the implementation and the complete demonstration of the experimental evaluation. Finally, the discussion and concluding statements are presented in Sections 6 and 7, respectively.

2. Related work

2.1. Low-light image enhancement

Low-light image enhancement aims to recover visually pleasant images from under-exposed inputs by correcting illumination while preserving structures and colors. Recent LIE approaches have significantly advanced restoration quality under fixed or known data distributions, including Transformer-based designs that leverage illumination-aware priors or Retinex-inspired formulations to improve enhancement fidelity (Cai et al., 2023a; Xu et al., 2022). However, many of these methods are primarily optimized for a single training distribution and may degrade when encountering illumination conditions unseen during training, where the learned mapping can overfit to dataset-specific lightness statistics.

A closely related line is generalized LIE, which explicitly targets robustness across varying illumination levels and capture conditions. CSNorm (Yao et al., 2023) is first designed to improve the generalization ability by utilizing the property of eliminating lightness-relevant components in normalization, and achieves satisfactory performance across different lightness levels. While effective in domain generalization settings, such methods typically ignore the intrinsic reasons for poor performance and neglect to consider the forgetting phenomenon in the continuous training process. In continual LIE, distribution shifts occur

over time and can exacerbate catastrophic forgetting. Hence, the challenge is not only to generalize across lightness levels but also to retain previously learned enhancement capability as new lightness conditions appear. This motivates our continual formulation and our Rehearsal-based Invariant Structure Regularizer, which maintains and transfers lightness-invariant knowledge across tasks.

2.2. Continual learning

Continual Learning entails an intelligent system's capacity to learn and accrue knowledge progressively without catastrophic forgetting. Typical CL scenarios include instance incremental learning, domain incremental learning, task incremental learning, and class incremental learning (Kong et al., 2023; Wang et al., 2024). Recent CL methods can also be divided into three categories based on how they prevent catastrophic forgetting: rehearsal-based, architecture-based, and regularization-based. The Rehearsal-based method, which is considered in this study, utilizes a memory buffer to store some data points or hidden representations from past tasks and replay them while training the current task (Liu et al., 2023; Wei et al., 2023; Yang et al., 2023b). Architecture-based methods accommodate multiple sub-networks within the CL model, thereby maintaining task isolation and mitigating the detrimental effects of catastrophic forgetting (Ksizaek & Spurek, 2025; Roy et al., 2024; Yu et al., 2024). Regularization-based methods explicitly add regularization terms to balance old and new tasks, which usually need a frozen copy of the model from former tasks (Cho et al. (2025), Kirkpatrick et al. (2017), Zenke et al. (2017)). Some works treat knowledge distillation as a distinct paradigm in CL taxonomies due to its prevalence and replay-free variants, where a previous model acts as a teacher to constrain the current model by matching outputs, features, or relational structures, often improving retention with limited or even no access to past data. Asadi et al. (2023), Hafez and Erekmek (2024), Phan et al. (2022). In our setting, we prioritize rehearsal-based CL over purely replay-free schemes because buffered exemplars offer stable supervision under sequential distribution shifts and better preserve the diversity of task-varying factors needed for robust continual enhancement.

Importantly, most mainstream CL research is developed under discriminative objectives (e.g., classification), whereas low-level vision tasks such as LIE involve dense prediction and require preserving image structures and photometric consistency. In such settings, forgetting may manifest as structural distortions, color shifts, or inconsistent illumination correction, which are not well captured by class-boundary preservation alone. Moreover the literature on CL has scarcely considered the effect of spurious correlations from the causal perspective, barring a few exceptions. (Jeon et al., 2023) observe that catastrophic forgetting is not always unfavorable and propose a method that encourages benign forgetting while maintaining useful information. (Lee et al., 2024) finds out that spurious correlations can be propagated across tasks, posing a significant challenge. Motivated by these gaps, we propose a unified framework that bridges continual learning for low-level vision with causal modeling to mitigate spurious correlations and improve robustness across sequential LIE tasks.

2.3. Causal inference

The causal inference is the process of discovering the causal relationship instead of correlation among multiple variables (Pearl et al., 2016). In recent years, there has been a growing trend of incorporating causal inference techniques into various computer vision tasks, including emotion recognition (Chen et al., 2022; Yang et al., 2023a), image segmentation (Li et al., 2024), and image classification (Sun et al., 2024; Tian et al., 2023). In the field of image restoration, Li et al. (2023b) pioneers a causal inference-based approach to image restoration, offering new insights into the complex relationships between degraded inputs and desired high-quality outputs.

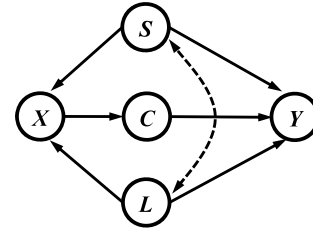


Fig. 2. The SCM diagram of the input low-light image X , model restored image Y , semantic information C , lightness condition L , and style information S .

Although the proposed distortion invariant representation learning paradigm in Li et al. (2023b) achieves impressive performance, some shortcomings remain that could be further investigated. Existing causal restoration efforts are typically developed under a static training regime and do not explicitly address the continual learning setting, where data distributions evolve over time. In continual low-light enhancement, illumination conditions and capture styles arrive sequentially, and task-varying factors can act as confounders that induce spurious correlations and may even be propagated through sequential updates. Therefore, it remains under-explored how to integrate causal modeling with continual low-level restoration to learn representations that are both robust to task-wise distribution shifts and resistant to catastrophic forgetting. Motivated by this gap, our work brings causal reasoning into continual LIE and encourages illumination-invariant representation learning under sequential task shifts.

3. Problem formulation

In this section, we model low-light image enhancement from the perspective of causal inference and analyze potential spurious correlations that may exist based on this modeling.

3.1. Preliminary

From a causality perspective, the process of low-light image enhancement can be described as follows. Given the low-light image X serving as the model input, we obtain the lightness-enhanced image Y by a neural network. In this process, several variables may get involved. These variables mainly include style information S and structural information C related to the X , as well as lightness conditions L .

3.2. Structural causal model

To comprehensively understand the intrinsic objective of LIE methods, which involves modeling invariant information from inputs under varying lightness conditions, we re-evaluate the learning paradigm of LIE through a causal framework and derive the corresponding SCM in Fig. 2. In this model, nodes represent abstract information variables, such as X , and directed edges denote functional causality; for example, $X \rightarrow C$ indicates that X serves as the cause and C as the effect. We subsequently provide an in-depth exposition of the SCM and the underlying rationale for its construction.

$X \rightarrow C \rightarrow Y$. The structural information of a low-light image should remain consistent with both the ground-truth image and the corresponding image after lightness adjustment. Therefore, the structural information C depends on the degraded image X and directly influences the lightness-adapted image Y .

$Y \leftarrow L \rightarrow X \leftarrow S \rightarrow Y$. Intuitively, the lightness conditions L directly impact the X and Y . Moreover, we assert that in addition to lightness conditions and image structural information independent of lightness, the style information of the image, including but not limited to color, also impacts the performance of low-light enhancement models. For example, low-light enhancement models should consider the impact of

color intensity on the effects of lightness enhancement. Compared to darker regions, the lighter areas of an image tend to exhibit more significant changes before and after lightness enhancement. Thus, S primarily encompasses image style information, such as color, can also influence both the low-light image X and the restored image Y .

$L \longleftrightarrow S$. The dashed arrow between the style information S and the lightness condition L in the SCM diagram indicates a spurious correlation. As we have discussed, the intensity of color affects the X and Y . However, the intensity of color can be attributed to both the intrinsic properties of the color itself as well as the influence of lightness. Similarly, the intensity of color can also interfere with the model's perception of lightness. Therefore, the causal relationship between S and L is unobservable, which can be considered as a spurious correlation. When the distribution of lightness is identical, this spurious correlation manifests as a relatively fixed bias. However, in a CL setting, the bias introduced by changing distributions can impact the model's generalization performance (Lee et al., 2024).

4. Methodology

In this section, to capture the desired causal effect for continual LIE, we delve into the derived SCM to determine the *confounding variables*. Then a novel *deconfounding* Rehearsal-based Invariant Structure Regularizer is proposed. In addition, the Channel Fourier Transform Self-Attention module is introduced for a precise estimation of causal effect.

4.1. Causal intervention via backdoor adjustment

Examining the SCM presented in Fig. 2, we can expound on the inherent objective of lightness adaptation from a causal perspective as follows: capturing the causal effect between the input degraded image X and the enhanced image Y , which is implemented by modeling $P(Y|X)$ in vanilla LIE methods. However, the unexpected backdoor paths among X and Y render a gap between $P(Y|X)$ and $P(Y|do(X))$. To disclose the focal issue hindering the development of existing approaches, we employ the interventionist theories of causation (Pearl et al., 2016) to analyze the SCM and reveal that Y is not only determined by X via the causal path, i.e., $X \rightarrow C \rightarrow Y$, but also affected by X via two *backdoor* paths¹ i.e., $X \leftarrow L \rightarrow Y$ and $X \leftarrow S \rightarrow Y$. The desired causal effect between the input image X and the enhanced image Y through the causal path $X \rightarrow C \rightarrow Y$ is hindered by the non-trivial interference stemming from two backdoor paths. Consequently, the conventional methods face difficulty in achieving robust lightness adaptation. In this context, L and S are considered as confounding factors with respect to accurately capturing the causal effect between X and Y . To address this issue, with the backdoor criterion (Pearl et al., 2016), we perform causal intervention via backdoor adjustment with the following formula:

$$P(Y = y|do(X = x)) = \sum_{s \in S} \sum_{l \in L} P(Y = y|L = l, S = s, X = x)P(L = l, S = s). \quad (1)$$

To this end, we propose to exclude the undesired effect introduced by the backdoor path from the estimation of the causal relationship between X and Y . *Intervention* (Shanmugam, 2001; Zelterman, 2001) is a relevant technique to block the undesired causal relationship by applying the *do* – calculus on the target variable. Adhering to the calculated backdoor adjustment formula, we can introduce causal intervention to sufficiently capture the desired causal effect of $X \rightarrow C \rightarrow Y$ by performing the probability expansion of L and S instead of modeling $P(Y = y|X = x)$. Due to the spurious correlation is eliminated by blocking L and S , modeling $P(Y|do(X))$ (rather than $P(Y|X)$) can avoid the bias introduced by the CL setting, thereby further improving the generalization capacity of the proposed CCLIE. To implement the Eq. (1),

we propose the Rehearsal-based Invariant Structure Regularizer module. It's worthy noting that, the backdoor adjustment aims to *estimate* the causal effect only based on the available samples without requiring additional experiments, such that the accurate estimation of the available value sets of the variables for adjustment widely determines the accuracy of estimating the desired causal effect. In our proposed SCM, the adequate value set of L, S is crucial to improve the estimation of the causal effect between X and Y , i.e., $P(Y|do(X))$. Considering that the complete value sets of lightness condition and style information are unavailable, we propose to estimate the lightness condition and style information from the input image X . For a precise estimation, we further introduce the Channel Fourier Transform-based Self-Attention module.

4.2. Rehearsal-based invariant structure regularizer

Based on the SCM analysis, we propose to substantially expand the available value set of the adjustment variables, i.e., S and L , in order to effectively implement the backdoor adjustment, which is achieved by introducing a counterfactual lightness augmentation approach. In the training process, the input data and the memory buffer data are two distinct sets of data with different structural, lightness, and style information. We can leverage the concept of Mixup (Zhang et al., 2018) to generate augmented data that maintains the same structure but exhibits varying lightness and style characteristics. From the causal inference perspective, since the structural information is preserved, the learning objective for the lightness adaptation task remains unchanged. However, the variations introduced in the S and L expand the available value set of the adjustment variables. This ensures that the target causal effect can be estimated more accurately through the backdoor adjustment.

To address the aforementioned issue, we need to first decompose an image into C, L , and S . Inspired by Zhou et al. (2023), we construct an augmented confounder dataset from the frequency domain perspective. When we obtain the frequency domain characteristics of images through the Fourier transform, the structural information is primarily contained within the phase component $\mathbf{P}$, while the lightness and style attributes reside in the amplitude component $\mathbf{A}$. Furthermore, since lightness is a global property, we can consider the low-frequency part of the amplitude component, $\mathbf{A}_{low}$, as a representation of the lightness information. Correspondingly, the high-frequency components of the amplitude, denoted as $\mathbf{A}_{high}$, can be taken to represent S .

$$\mathbf{A} = \mathcal{A}(\mathcal{F}(X)), \mathbf{P} = \mathcal{P}(\mathcal{F}(X)), \quad (2a)$$

$$\mathbf{A}_{low} = \mathbf{A}(0, 0), \mathbf{A}_{high} = \mathbf{A} \setminus \mathbf{A}_{low}. \quad (2b)$$

Therefore, our augmentation process can be described as follows:

$$\hat{\mathbf{A}} = \mathbf{A}_{inp,low} \times \alpha + \mathbf{A}_{buf,low} \times (1 - \alpha) + \mathbf{A}_{inp,high} \times \beta + \mathbf{A}_{buf,high} \times (1 - \beta), \quad (3a)$$

$$\mathbf{R} = \hat{\mathbf{A}} \times \cos(\mathbf{P}_{inp}), \mathbf{I} = \hat{\mathbf{A}} \times \sin(\mathbf{P}_{inp}), \quad (3b)$$

$$\hat{X} = \mathcal{F}^{-1}(\mathbf{R} + j\mathbf{I}). \quad (3c)$$

Specifically, we convert the input X into the frequency representation through the Fourier transform $\mathcal{F}$ and obtain the amplitude component $\mathbf{A}$ and phase component $\mathbf{P}$. The subscripts 'inp' and 'buf' denote whether the image originates from the input data or the memory buffer data, respectively. α and β are random variables uniformly distributed over the interval $[0, 1]$. Finally, we convert the real part $\mathbf{R}$ and the imaginary part $\mathbf{I}$ to the spatial domain and obtain the augmentation data $\hat{X}$.

During training, we feed $\hat{X}$ into the model, with both buffer data and input data. The resulting loss term serves as the Rehearsal-based Invariant Structure Regularizer, denoted as L_{RISR} . Due to the inherent difficulty in perfectly separating these three variables through the frequency domain-based decoupling approach, the obtained $\hat{X}$ may introduce undesirable artifacts, which could have a detrimental effect on

¹ In SCM, a backdoor path causes an *unordered dependence* (Li et al., 2023a) between the end-side variables, which is non-causality.

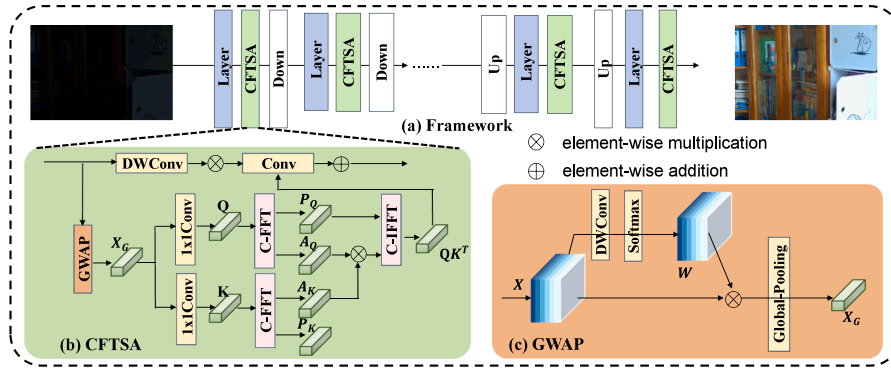


Fig. 3. Illustration of the proposed framework and its components. (a) The structure of the Convolutional Fourier Transform Self-Attention (CFTSA) module, which combines Fourier transform operations and self-attention mechanisms to effectively capture global and local dependencies. (b) The design of the Global Weighted Average Pooling (GWAP) module in CFTSA which calculates global features by aggregating spatial information through attention weights. (c) The positions where the proposed plug-and-play CFTSA module is inserted into the existing model. For simplicity, we omit residual connections and some repetitive modules.

model performance. While maintaining the beneficial effect of expanding the available value set, we assign a relatively small weight λ to the regularization term L_{RISR} , in order to achieve a balance between the two objectives. Thus, the total loss function can be presented as:

$$L_{total} = L_{ori} + \lambda \cdot L_{RISR}, \quad (4)$$

L_{ori} represents the loss function computed from the input data and the buffer data.

4.3. Channel Fourier transform-based self-attention

The self-attention mechanism selectively enhances and suppresses the weights of different parts of the input data, thereby focusing the network's attention on the most salient features. To enable the network to focus on low-frequency and high-frequency information in different parts of the features, we revisit the self-attention mechanism and propose a plug-and-play module in this section. Considering that GAP and low-pass filtering exhibit similar characteristics (Cui et al., 2023b; Qin et al., 2021), we utilize a novel Global Weighted Average Pooling to substitute the QK projection layer in the self-attention mechanism and calculate *dot product* $Q \cdot K^T$ in the frequency domain, as shown in Fig. 3 (a). This enables the network to learn more effective representations of low-frequency information, which is related to lightness conditions in this study, thereby improving the generalization to diverse lightness conditions.

Global Weighted Average Pooling. Traditional pooling methods, such as Global Average Pooling (GAP), treat all features equally, neglecting the importance of focusing on discriminative representations. To address this limitation, we propose Global Weighted Average Pooling (GWAP), which adaptively selects features with corresponding weights, thereby enhancing the ability to learn critical representations for tasks like reconstruction. As shown in Fig. 3 (b), GWAP operates on a feature tensor $x \in \mathbb{R}^{H \times W \times C}$, where H , W , and C represent the height, width, and channel. First, x is reshaped to $\mathbb{R}^{HW \times C}$, and a Softmax function is applied across the spatial dimensions to compute a global weight descriptor. This weight descriptor is reshaped back to its original shape, yielding a weight map $w \in \mathbb{R}^{H \times W \times C}$. The input features are then activated by the weight map through element-wise multiplication, and the resulting features are summed along the spatial dimensions, producing a global weighted vector x_g with a shape of $1 \times 1 \times C$. The process is formulated as:

$$x_g = \text{sum}(w \otimes x), \quad (5)$$

where $\otimes$ denotes element-wise multiplication and sum represents the sum operation along the spatial dimension. Unlike GAP, GWAP introduces a weighting mechanism through a parallel branch that computes weights using depth-wise convolution followed by Softmax activation,

ensuring the sum of weights across spatial dimensions for each channel equals one. This adaptivity allows GWAP to improve discriminative capacity and produce generalized feature representations. Consequently, the weighted features are directly summed, enabling efficient dimensionality reduction while maintaining critical spatial information for subsequent processing.

Plug-and-Play. Considering that existing low-light image enhancement networks are typically constructed with an encoder-decoder architecture and incorporate a multi-scale feature extraction design, the encoder or decoder is often composed of sequential feature layers connected with downsampling or upsampling layers. The CFTSA module is strategically inserted between the primary layers of the encoder and decoder to enhance feature representation and capture critical information, as illustrated in Fig. 3 (a). This integration of GWAP and frequency domain operations enables the network to effectively learn representations of low-frequency information, which is closely related to lightness conditions in this study. By improving the focus on both low-frequency and high-frequency components, the proposed module enhances generalization across diverse lightness conditions, thereby making the model more robust and effective in real-world scenarios.

5. Experiments

In this section, we perform experiments on multiple baselines and datasets to thoroughly assess the effectiveness of our approach. Furthermore, experiments are conducted to evaluate the improvement in generalization ability through cross-dataset testing and downstream object detection tasks. Additionally, we conduct ablation studies to assess the contribution of each component of CCLIE.

5.1. Implementation details

We evaluate our method on LOL (Wei et al., 2018a) and Huawei (Hai et al., 2023) datasets. The LOL dataset is a widely used benchmark for low-light image enhancement research. It consists of 485 real-world image pairs for training and 15 pairs for testing. The Huawei dataset is a large-scale real-world dataset and contains 2480 paired images taken by the HUAWEI P40 Pro mobile phone. The training and testing set of the Huawei dataset is split in the proportion of 2450:30.

Given the difficulty in acquiring real-world images under diverse lightness conditions, we adopt the method proposed in Wang et al. (2023) to generate synthetic datasets. We use the adjustment coefficient α to represent datasets with different lightness distributions, where $\alpha < 1$ corresponds to lower lightness and $\alpha > 1$ corresponds to higher lightness. We obtained two synthetic datasets with $\alpha = 0.5$ and $\alpha = 2.0$ for continuous training. The training sequence is conducted in the order of $\alpha = 0.5$, $\alpha = 2.0$, and the original LOL and Huawei dataset. Additionally,

Table 1

Quantitative comparisons on LOL and Huawei datasets, where *original* represents the original lightness levels ($\alpha = 1.0$). The $\alpha = 0.4$ and $\alpha = 0.5$ denote the darker images than *original* images, and the $\alpha = 2.0$ and $\alpha = 2.4$ denote the brighter ones. underline+ and *italic*0 denote the underlineimprovement and *italic*decline of the PSNR metrics.

Methods	LOL						Huawei					
	original	$\alpha = 0.4$	$\alpha = 2.4$	$\alpha = 0.5$	$\alpha = 2$	Avg.	original	$\alpha = 0.4$	$\alpha = 2.4$	$\alpha = 0.5$	$\alpha = 2$	Avg.
LIME	14.22	11.28	17.19	11.78	16.68	14.23	15.34	12.42	17.52	13.09	17.72	15.22
RetinexNet	16.77	16.11	15.61	16.44	15.73	16.13	18.59	16.09	15.29	17.12	16.38	16.69
ZeroDCE	14.80	17.66	10.98	17.53	11.72	14.54	15.14	17.69	10.67	17.74	11.55	14.56
SCI	14.78	9.41	15.49	10.35	17.00	13.41	15.70	10.77	15.24	11.74	16.41	13.97
FourLLIE	21.91	17.75	18.97	17.60	20.69	19.38	21.30	18.67	18.43	19.58	19.39	19.47
DRBN	20.21	17.37	17.40	18.17	18.26	18.28	20.05	18.62	18.36	19.24	19.04	19.06
DRBN-Seq	19.76	16.88	17.79	17.55	18.31	18.06	19.13	19.88	13.36	19.99	14.60	17.39
DRBN-CCLIE	21.28	20.01	20.30	20.92	20.58	20.62	20.96	20.05	20.64	20.61	20.60	20.57
<i>Gains</i>	<u>+1.52</u>	<u>+3.13</u>	<u>+2.51</u>	<u>+3.37</u>	<u>+2.27</u>	<u>+2.56</u>	<u>+1.83</u>	<u>+0.17</u>	<u>+7.28</u>	<u>+0.62</u>	<u>+6.00</u>	<u>+3.18</u>
DRBN-Mix	19.42	18.75	19.68	19.32	19.44	19.32	20.28	19.55	20.34	19.53	20.29	20.00
RetinexFormer	23.42	20.66	16.47	20.80	17.65	19.80	20.90	19.46	16.69	20.25	17.83	19.03
RetinexFormer-Seq	23.25	20.82	21.79	23.01	22.59	22.29	21.19	18.08	16.83	19.09	18.27	18.69
RetinexFormer-CCLIE	23.63	21.26	22.54	24.19	22.83	22.89	21.11	20.30	20.12	20.42	20.74	20.54
<i>Gains</i>	<u>+0.38</u>	<u>+0.44</u>	<u>+0.75</u>	<u>+1.18</u>	<u>+0.24</u>	<u>+0.60</u>	<i>-0.08</i>	<u>+2.22</u>	<u>+3.29</u>	<u>+1.33</u>	<u>+2.47</u>	<u>+1.85</u>
RetinexFormer-Mix	23.08	23.19	23.13	23.10	23.01	23.10	21.12	20.78	20.36	20.74	20.99	20.80
FocalNet	23.59	19.47	18.96	20.67	20.17	20.57	21.20	17.63	18.23	19.05	19.07	19.04
FocalNet-Seq	23.70	18.81	18.54	20.57	20.26	20.38	21.23	20.25	19.28	20.51	19.82	20.22
FocalNet-CCLIE	24.44	22.43	23.96	23.61	23.69	23.63	<u>21.24</u>	20.79	20.39	20.91	20.72	20.81
<i>Gains</i>	<u>+0.74</u>	<u>+3.62</u>	<u>+5.42</u>	<u>+3.04</u>	<u>+3.43</u>	<u>+3.25</u>	<u>+0.01</u>	<u>+0.54</u>	<u>+1.11</u>	<u>+0.40</u>	<u>+0.90</u>	<u>+0.59</u>
FocalNet-Mix	23.63	23.59	23.57	23.55	23.54	23.58	21.17	20.87	21.36	20.95	21.30	21.13

we also create test sets with $\alpha = 0.4$ and $\alpha = 2.4$ to evaluate the model's generalization performance on unseen data.

To validate the effectiveness of the proposed method, we compare it with other state-of-the-art approaches, including LIME (Guo et al., 2016), RetinexNet (Wei et al., 2018b), ZeroDCE (Guo et al., 2020), SCI (Ma et al., 2022), FourLLIE (Wang et al., 2023), and Retinexformer (Cai et al., 2023b). Furthermore, we select two representative methods as baseline methods and integrate our method into them, including the latest FocalNet (Cui et al., 2023c) and the widely used baseline network DRBN (Yang et al., 2020), to show the effectiveness in alleviating catastrophic forgetting and superior generalization performance of our method. The memory buffer mechanism in our method is introduced from ER (Chaudhry et al., 2019).

The experiments are conducted with PyTorch on NVIDIA Tesla V100 GPUs. We keep all the training settings consistent with the default settings of baseline methods. While the training epoch for the first task remains consistent, the training epoch for subsequent tasks is reduced by half. For our method, we set the weight $\lambda = 0.001$ for RISR and memory buffer size as 50 and 200 for the LOL dataset and Huawei dataset, respectively. The suffix "Seq" denotes that the model was sequentially trained on datasets with varying lightness conditions, without employing any CL methods. Conversely, the suffix "Mix" indicates that the model was trained on a comprehensive training set that encompassed all the lightness conditions. Regarding evaluation, we choose the Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity (SSIM) as the evaluation metrics to validate the performance of different methods.

5.2. Experimental results and analysis

Quantitative and qualitative results are reported in Table 1 and Figs. 4, 5, 6 respectively. Despite the promising performance of some recent approaches, such as FocalNet and RetinexFormer, on original benchmark datasets, a significant drop in performance has been observed when these methods are evaluated on unseen data ($\alpha = 0.5$ and $\alpha = 2.4$). By incorporating our proposed approach, significant performance improvements have been observed for both FocalNet and DRBN across diverse lightness conditions in the evaluation tests. The observed improvements in the average PSNR metric indicate that our proposed approach enhances the model's ability to generalize across diverse lightness conditions. Furthermore, by training the model under a

CL paradigm and incorporating causal backdoor adjustment, the model demonstrates an enhanced ability to effectively capture the causal relationship between the input degraded image and the restored output. This leads to improved performance on the most recent task, which in this scenario corresponds to the original dataset ($\alpha = 1.0$).

Comparison with sequential training. To ensure the fairness of the experimental evaluation and verify that the observed performance improvements are not simply attributed to increased training data, we conducted sequential training, reported as FocalNet-Seq and DRBN-Seq in Table 1. In contrast to the baseline, the sequential variants DRBN-Seq and FocalNet-Seq both exhibited performance degradation, with the former on two distinct datasets and the latter specifically on the LOL dataset. This observation reflects the potential for individual dataset biases to exert adverse impacts on the performance of both the former and subsequent tasks. However, after incorporating our proposed approach, FocalNet demonstrates performance improvements of 3.25dB and 0.59dB of PSNR on the LOL and Huawei datasets, respectively. Concurrently, DRBN exhibits enhancements of 2.56dB and 3.18dB.

We also provide additional qualitative results for FocalNet before and after the incorporation of our proposed CCLIE in Fig. 7. The results demonstrate that our method exhibits strong generalization performance, effectively preserving color consistency under varying lighting conditions.

Comparison with mixing training. The DRBN-Ours significantly outperforms DRBN-Mix across both evaluated datasets. In contrast, FocalNet-CCLIE only exhibits a slight performance advantage over FocalNet-Mix, specifically on the LOL dataset. Intuitively, the training paradigm that involves the aggregation of all available datasets tends to exhibit the best overall performance. This observed performance enhancement suggests that our proposed plug-and-play module and backdoor adjustment technique can more effectively uncover the true causal effects to facilitate generalized low-light image enhancement. This further suggests that a deeper investigation of low-light enhancement techniques within the context of CL approaches would be a promising avenue.

Downstream Object Detection. We perform low-light object detection experiments on the ExDark dataset (Loh & Chan, 2019), as presented in Table 2. A pre-trained YOLO-v3 detector is employed to evaluate the performance across three sets of images: the original low-light source images, the images enhanced by the backbone network model,

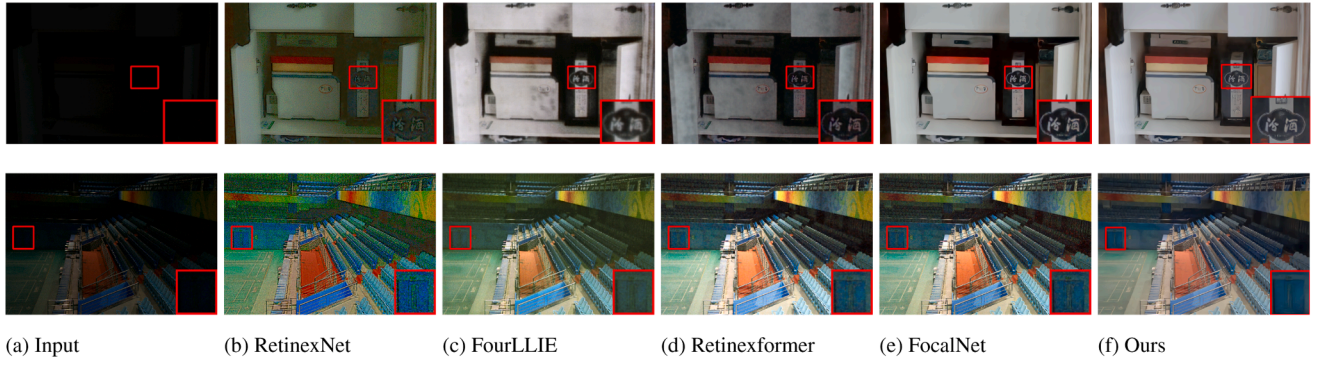


Fig. 4. Qualitative results of different methods on LOL dataset. The first row is the results of $\alpha = 0.4$ and the second row is the results of $\alpha = 2.4$. Our method shows superior performance in alleviating artifacts, suppressing noise and preserving color consistency.



Fig. 5. More qualitative results of different methods on darker LOL dataset ($\alpha = 0.4$).

Table 2

Comparison of methods on object detection tasks.

	Bic.	Boat	Bottle	Bus	Car	Cat	Chair	Cup	Dog	Mot.	Peo.	Tab	mAP
Source	0.586	0.430	0.431	0.723	0.542	0.376	0.363	0.396	0.490	0.351	0.413	0.306	0.451
RetinexFormer w/ CCLIE	0.667 0.680	0.505 0.490	0.563 0.592	0.727 0.741	0.568 0.568	0.501 0.476	0.499 0.486	0.432 0.434	0.615 0.618	0.383 0.391	0.516 0.524	0.387 0.380	0.530 0.532
FocalNet w/ CCLIE	0.685 0.680	0.512 0.527	0.588 0.591	0.748 0.759	0.573 0.583	0.500 0.520	0.519 0.519	0.446 0.485	0.627 0.630	0.420 0.429	0.550 0.533	0.410 0.402	0.548 0.555

and the images enhanced by the backbone model integrated with our proposed method CCLIE. The mean Average Precision (mAP) is compared among these groups to assess the effectiveness of different enhancement techniques. The incorporation of the CCLIE module consistently improves the performance of both FocalNet and RetinexFormer across most categories, demonstrating its effectiveness in enhancing object detection capabilities. The bold values highlight the best results for each category, indicating that models with CCLIE achieve higher scores. With the growing interest in image enhancement and restoration methods that target in-the-wild scenarios, we aim to extend our research in the future to explore further advancements in this domain.

Generalization Performance. We experiment with generalization performance across different datasets in Table 3. We train our model on the Huawei and test it on the LOL, and vice versa. Previous approaches

often demonstrate a tendency to overfit the training dataset, resulting in limited generalization capabilities. For each method, the incorporation of CCLIE consistently improves performance across both cross-dataset settings (Huawei / LOL and LOL / Huawei), as indicated in Table 3. Our CCLIE method improves the performance of baseline networks, which greatly enhances their generalization abilities in unknown lightness conditions.

5.3. Complexity analysis

We further investigate the computational complexity increased by the proposed CFTSA and GWAP. In practice, we incorporated three CFTSA modules into both the encoder and decoder sections of the baseline methods. Table 4 presents a computational complexity comparison



Fig. 6. More qualitative results of different methods on brighter LOL dataset ($\alpha = 2.4$).

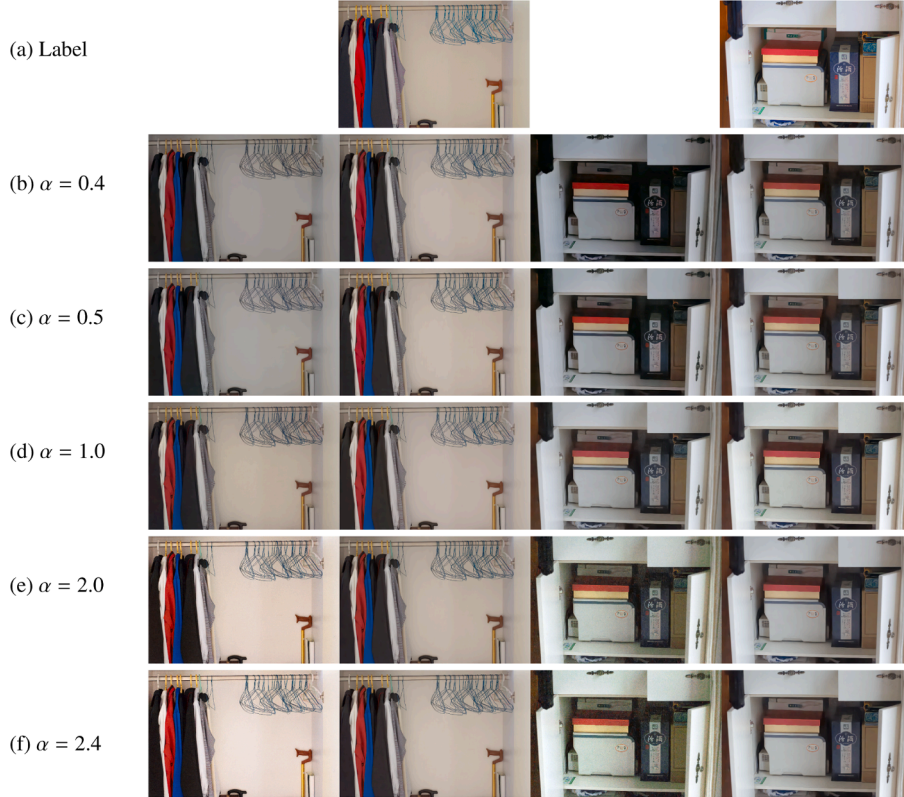


Fig. 7. More comparisons between the FocalNet and our method on the LOL dataset. The first and third columns regard FocalNet, while the second and fourth columns correspond to our proposed method. (a) shows the ground-truth image, and (b) to (f) depict inputs with varying lightness conditions.

between the proposed methods and baseline methods, focusing on two key metrics: FLOPs (FLOPs/G) and Parameters (Params/M). For FocalNet, the FLOPs/G is 111.82, which increases to 112.84 with the addition of CCLIE, resulting in a change of +1.02 (+0.91%). In terms of parameters, FocalNet has a count of 3.74 million, rising to 3.79 million with CCLIE, reflecting an increase of +0.05 million (+1.3%). For DRBN, the FLOPs start at 58.25 and rise to 59.46 with CCLIE, leading to a change of +1.21 (+2.08%). The parameter count for DRBN is 0.558 million,

increasing to 0.576 million with CCLIE, resulting in a change of +0.018 million (+3.2%). Overall, the integration of CCLIE leads to a slight increase in both computational complexity and parameter count for both methods, with DRBN exhibiting a higher percentage increase in both metrics compared to FocalNet.

As shown in Table 4, integrating CCLIE introduces only a marginal increase in FLOPs and parameters, indicating that the inference-time overhead is limited. In contrast, the main efficiency trade-off arises

Table 3

Comparison of methods on Huawei/LOL and LOL/Huawei datasets.

Train / Test	Huawei / LOL	LOL / Huawei
CLANE	10.25 / 0.5602	11.11 / 0.4152
LIME	15.25 / 0.7994	15.20 / 0.5321
RetinexNet	15.35 / 0.5102	15.32 / 0.4855
ZeroDCE	15.01 / 0.5974	12.25 / 0.4194
DRBN	18.10 / 0.8033	15.41 / 0.7047
DRBN-CCLIE	18.68 / 0.8058	17.32 / 0.7507
RetinexFormer	17.59 / 0.7846	17.58 / 0.7561
RetinexFormer-CCLIE	17.92 / 0.8033	18.29 / 0.7753
FocalNet	18.19 / 0.7784	17.96 / 0.8020
FocalNet-CCLIE	18.60 / 0.7781	19.19 / 0.8101

Table 4

Computational complexity comparison of our proposed method and baseline methods.

Methods	FLOPs/G	Params/M
DRBN	58.25	0.558
DRBN + CCLIE	59.46	0.576
Δ	+ 1.21 (2.08%)	+ 0.018 (3.2%)
FocalNet	111.82	3.74
FocalNet + CCLIE	112.84	3.79
Δ	+ 1.02 (0.91%)	+ 0.05 (1.3%)

Table 5

Ablation study for each component contribution evaluated in terms of PSNR / SSIM regarding LOL dataset.

Core Components	original		Avg.	
	PSNR $\uparrow$	SSIM $\uparrow$	PSNR $\uparrow$	SSIM $\uparrow$
FocalNet(Baseline)	23.59	0.9095	20.57	0.8745
+ ER	23.72	0.9155	21.95	0.9033
+ RISR	23.20	0.9044	22.70	0.9017
+ CFTSA	24.44	0.9102	23.63	0.9093

during training in the continual setting: as the number of sequential low-light tasks increases, the overall training time scales approximately linearly with the number of tasks under a fixed training schedule per task, and the rehearsal mechanism introduces a memory-compute trade-off. Specifically, with a fixed global buffer size, the exemplar budget per task shrinks as the number of tasks increases, potentially weakening coverage of past data distributions; conversely, enforcing a fixed per-task buffer allocation yields an approximately linear growth in total memory with respect to the number of tasks. In addition, increasing the buffer capacity or replay ratio typically improves retention and generalization at the expense of higher training-time computation. We further analyze the impact of the memory buffer size in the subsequent ablation study.

5.4. Ablation studies

Ablation on the core components. Our method consists of three components: rehearsal-based CL framework ER, rehearsal-based invariant structure regularizer, and channel Fourier Transform-based self-attention. We analyze the effect of each aspect in Table 5. We could discover that although RISR yields some gains on average metrics by expanding the available value set of the adjustment variables, it brings artifacts to enhanced images in the latest task. This observation further demonstrates the necessity of disentangling these variables within the feature representations in order to enable a more accurate estimation of the causal effects. To achieve this, as shown in the table, the Channel Fourier Transform-based Self-Attention mechanism we have proposed further enhances the model's performance.

Table 6Ablation study on the weight λ of the rehearsal-based invariant structure regularizer regarding the LOL dataset.

L_{RISR} weight λ	original		Avg.	
	PSNR $\uparrow$	SSIM $\uparrow$	PSNR $\uparrow$	SSIM $\uparrow$
0.5	23.06	0.9002	22.84	0.8984
0.1	23.41	0.9095	23.30	0.9087
0.01	24.44	0.9102	23.63	0.9093
0.001	<u>23.44</u>	0.9110	<u>23.37</u>	0.9114

Table 7

Ablation study on the memory buffer size of the proposed method regarding the Huawei dataset.

Buffer Size	original		Avg.	
	PSNR $\uparrow$	SSIM $\uparrow$	PSNR $\uparrow$	SSIM $\uparrow$
50	20.81	0.8124	20.06	0.7912
100	20.96	0.8168	20.33	0.7934
200	21.24	0.8191	20.81	0.8075

Ablation on the weight of RISR. We also conduct an ablation study on the weight λ of RISR. The result is reported in Table 6. Smaller weighting values can mitigate the adverse impact of artifacts introduced by the RISR. However, excessively small weighting values would render the RISR mechanism ineffective. Consequently, we selected a weighting value of 0.01 as a balance between minimizing artifacts and preserving the efficacy of the RISR.

Ablation on the memory buffer size. Finally, we experimented with the memory buffer size in Table 7. The results demonstrate that as the buffer size is increased, the model exhibits a significant improvement in performance on both the latest task as well as the average across all test sets. Conversely, an excessively small buffer size leads to overfitting of the data stored for previous tasks, resulting in a degradation of the model's overall performance. However, an overly large buffer size can substantially increase the computational burden on the model. For the Huawei dataset, which contains 2450 training samples, a buffer size of 200 would require storing over 8% of the total training data. Accordingly, we selected buffer sizes of 50 and 200 for the LOL and Huawei datasets, respectively, taking into account the scale of each dataset.

6. Discussion

The experimental results demonstrate that our proposed method significantly improves generalization performance compared to the baselines and achieves superior performance on the test set relative to the mixed-data paradigm. In the context of low-light image enhancement, the introduction of a continual learning framework presents both challenges, such as catastrophic forgetting, and opportunities for performance improvement. Unlike traditional continual learning tasks, we observe that the domain differences between sequential tasks in low-light enhancement are relatively smaller. This characteristic can be leveraged to improve generalization to unseen lighting conditions that are not drastically different from the training data. However, existing low-light enhancement methods often overlook these patterns, resulting in sub-optimal performance. Therefore, it is essential to effectively utilize these domain-specific characteristics to further enhance model performance, ensuring greater robustness and applicability in real-world scenarios.

Limitations and Future work. In the generalization experiments, although our method demonstrates performance gains in downstream task evaluations, the improvements remain limited. Through analysis, we hypothesize that this limitation may stem from incomplete decoupling in the frequency domain, which introduces additional artifacts and constrains the network's enhancement capabilities. To address this issue, future research can explore two potential directions. First, incorporating physical priors into the image decoupling process could provide

more robust guidance and reduce artifacts. Second, designing additional learnable modules could enable the network to adaptively extract relevant features for better enhancement. These approaches have the potential to further improve the generalization ability and effectiveness of the proposed framework.

7. Conclusion

In this paper, we investigate the challenging continual low-light image enhancement for the first time. Through SCM-based theoretical analysis, we present a plug-and-play Continual Low-light Image Enhancement framework, dubbed CCLIE. In CCLIE, we introduce the Rehearsal-based Invariant Structure Regularizer to address the inherent confounding effects and propose a Channel Fourier Transform-based Self-Attention to estimate a more precise causal effect. Extensive experiments demonstrate that CCLIE can alleviate catastrophic forgetting and effectively improve the performance of the baseline methods on LIE benchmarks. This result further suggests that the low-light image enhancement task can potentially benefit from continual learning frameworks. However, existing continual learning approaches have primarily focused on discriminative tasks. Methods for integrating continual learning with low-level tasks warrant further exploration, which we intend to address in our future research.

CRedit authorship contribution statement

Fan Ji: Writing – original draft, Methodology, Investigation, Conceptualization; **Hao Li:** Investigation, Data curation; **Jiangmeng Li:** Writing – review & editing, Supervision, Formal analysis; **Xiongxin Tang:** Writing – review & editing; **Fanjiang Xu:** Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

Asadi, N., Davari, M., Mudur, S., Aljundi, R., & Belilovsky, E. (2023). Prototype-sample relation distillation: Towards replay-free continual learning. In *Proceedings of the 40th international conference on machine learning ICML'23*. JMLR.org.

Cai, Y., Bian, H., Lin, J., Wang, H., Timofte, R., & Zhang, Y. (2023a). Retinexformer: One-stage retinex-based transformer for low-light image enhancement. In *Proceedings of the IEEE/CVF international conference on computer vision* (pp. 12504–12513).

Cai, Y., Bian, H., Lin, J., Wang, H., Timofte, R., & Zhang, Y. (2023b). Retinexformer: One-stage retinex-based transformer for low-light image enhancement. In *2023 IEEE/CVF International conference on computer vision (ICCV)* (pp. 12470–12479). Los Alamitos, CA, USA: IEEE Computer Society.

Chaudhry, A., Rohrbach, M., Elhoseiny, M., Ajanthan, T., Dokania, P., Torr, P., & Ranzato, M. (2019). Continual learning with tiny episodic memories. In *Workshop on multi-task and lifelong reinforcement learning*.

Chen, Y., Yang, X., Cham, T.-J., & Cai, J. (2022). Towards unbiased visual emotion recognition via causal intervention. In *Proceedings of the 30th ACM international conference on multimedia* (pp. 60–69).

Cho, S., Lee, H., Baek, S., & Paik, S.-B. (2025). Neuromimetic metaplasticity for adaptive continual learning without catastrophic forgetting. *Neural Networks*, 190, 107762.

Cui, Y., Ren, W., Cao, X., & Knoll, A. (2023a). Focal network for image restoration. In *2023 IEEE/CVF International conference on computer vision (ICCV)* (pp. 12955–12965).

Cui, Y., Tao, Y., Bing, Z., Ren, W., Gao, X., Cao, X., Huang, K., & Knoll, A. (2023b). Selective frequency network for image restoration. In *The eleventh international conference on learning representations*.

Cui, Y., Tao, Y., Bing, Z., Ren, W., Gao, X., Cao, X., Huang, K., & Knoll, A. (2023c). Selective frequency network for image restoration. In *The eleventh international conference on learning representations*.

Cui, Y., Zamir, S. W., Khan, S., Knoll, A., Shah, M., & Khan, F. S. (2024). AdaIR: Adaptive all-in-one image restoration via frequency mining and modulation. arXiv preprint arXiv:2403.14614.

Guo, C. G., Li, C., Guo, J., Loy, C. C., Hou, J., Kwong, S., & Cong, R. (2020). Zero-reference deep curve estimation for low-light image enhancement. In *Proceedings of the IEEE conference on computer vision and pattern recognition (CVPR)* (pp. 1780–1789).

Guo, X., & Hu, Q. (2023). Low-light image enhancement via breaking down the darkness. *International Journal of Computer Vision*, 131(1), 48–66.

Guo, X., Li, Y., & Ling, H. (2016). Lime: Low-light image enhancement via illumination map estimation. *IEEE Transactions on Image Processing*, 26(2), 982–993.

Hafez, M. B., & Erekmen, K. (2024). Continual deep reinforcement learning with task-agnostic policy distillation. *Scientific Reports*, 14(1), 31661.

Hai, J., Xuan, Z., Yang, R., Hao, Y., Zou, F., Lin, F., & Han, S. (2023). R2RNet: Low-light image enhancement via real-low to real-normal network. *J. Vis. Commun. Image Represent.*, 90(C), 103712.

He, R., Li, X., & Wu, J. (2025). LeeSDFormer: a lightweight unsupervised cnn-transformer-based curve estimation network for low-light image enhancement, exposure suppression, and denoising. *Neural Networks*, 190, 107764.

Jeon, M., Lee, H., Seong, Y., & Kang, M. (2023). Learning without prejudices: Continual unbiased learning via benign and malignant forgetting. In *The eleventh international conference on learning representations*.

Kirkpatrick, J., Pascanu, R., Rabinowitz, N., Veness, J., Desjardins, G., Rusu, A. A., Milan, K., Quan, J., Ramalho, T., Grabska-Barwinska, A., Hassabis, D., Clopath, C., Kumaran, D., & Hadsell, R. (2017). Overcoming catastrophic forgetting in neural networks. *Proceedings of the National Academy of Sciences*, 114(13), 3521–3526.

Kong, Y., Liu, L., Qiao, M., Wang, Z., & Tao, D. (2023). Trust-region adaptive frequency for online continual learning. *International Journal of Computer Vision*, 131(7), 1825–1839.

Ksiazek, K., & Spurek, P. (2025). Hypermask: Adaptive hypernetwork-based masks for continual learning. *Neural Networks*, 191, 107736.

Lee, D., Jung, S., & Moon, T. (2024). Continual learning in the presence of spurious correlations: Analyses and a simple baseline. In *The twelfth international conference on learning representations*.

Li, C., Mao, Y., Liang, S., Li, J., Wang, Y., & Guo, Y. (2024). Deep causal learning for pancreatic cancer segmentation in CT sequences. *Neural Networks*, 175, 106294.

Li, J., Zhang, Y., Qiang, W., Si, L., Jiao, C., Hu, X., Zheng, C., & Sun, F. (2023a). Disentangle and remerge: Interventional knowledge distillation for few-shot object detection from a conditional causal perspective. In *Proceedings of the thirty-seventh AAAI conference on artificial intelligence and thirty-fifth conference on innovative applications of artificial intelligence and thirteenth symposium on educational advances in artificial intelligence AAAI'23/IAAI'23/EAAI'23*. AAAI Press.

Li, X., Li, B., Jin, X., Lan, C., & Chen, Z. (2023b). Learning distortion invariant representation for image restoration from a causality perspective. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 1714–1724).

Liu, Y., Cong, Y., Goswami, D., Liu, X., & van de Weijer, J. (2023). Augmented box replay: Overcoming foreground shift for incremental object detection. In *2023 IEEE/CVF International conference on computer vision (ICCV)* (pp. 11333–11343). Los Alamitos, CA, USA: IEEE Computer Society.

Loh, Y. P., & Chan, C. S. (2019). Getting to know low-light images with the exclusively dark dataset. *Computer Vision and Image Understanding*, 178, 30–42.

Lv, F., Li, Y., & Lu, F. (2021). Attention guided low-light image enhancement with a large scale low-light simulation dataset. *International Journal of Computer Vision*, 129(7), 2175–2193.

Ma, L., Ma, T., Liu, R., Fan, X., & Luo, Z. (2022). Toward fast, flexible, and robust low-light image enhancement. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 5637–5646).

McCloskey, M., & Cohen, N. J. (1989). Catastrophic interference in connectionist networks: The sequential learning problem. In G. H. Bower (Ed.), *Psychology of learning and motivation* (pp. 109–165). Academic Press (vol. 24). Psychology of Learning and Motivation.

Pearl, J., Glymour, M., & Jewell, N. P. (2016). Causal inference in statistics: A primer. 2016. *Internet Resource*.

Phan, M. H., Ta, T.-A., Phung, S. L., Tran-Thanh, L., & Bouzerdoum, A. (2022). Class similarity weighted knowledge distillation for continual semantic segmentation. In *2022 IEEE/CVF Conference on computer vision and pattern recognition (CVPR)* (pp. 16845–16854). <https://doi.org/10.1109/CVPR52688.2022.01636>

Qin, Z., Zhang, P., Wu, F., & Li, X. (2021). Fcanet: Frequency channel attention networks. In *2021 IEEE/CVF International conference on computer vision (ICCV)* (pp. 763–772).

Roy, A., Moullick, R., Verma, V. K., Ghosh, S., & Das, A. (2024). Convolutional prompting meets language models for continual learning. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 23616–23626).

Shanmugam, R. (2001). Causality: Models, reasoning, and inference: Judea pearl; cambridge university press, cambridge, uk, 2000, pp 384, ISBN 0-521-77362-8. *Neurocomputing*, 41(1-4), 189–190.

Sun, H., Wei, Q., Feng, L., Hu, Y., Liu, F., Fan, H., & Yin, Y. (2024). Variational rectification inference for learning with noisy labels. *International Journal of Computer Vision*, (pp. 1–20).

Tian, Y., Bai, K., Yu, X., & Zhu, S. (2023). Causal multi-label learning for image classification. *Neural Networks*, 167, 626–637.

Wang, C., Wu, H., & Jin, Z. (2023). FourLLIE: Boosting low-light image enhancement by fourier frequency information. In *Proceedings of the 31st ACM international conference on multimedia MM '23* (p. 7459-7469). New York, NY, USA: Association for Computing Machinery.

- Wang, L., Zhang, X., Su, H., & Zhu, J. (2024). A comprehensive survey of continual learning: Theory, method and application. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 46(8), 5362–5383.
- Wei, C., Wang, W., Yang, W., & Liu, J. (2018a). Deep retinex decomposition for low-light enhancement. *arXiv preprint arXiv:1808.04560*.
- Wei, C., Wang, W., Yang, W. Y., & Liu, J. (2018b). Deep retinex decomposition for low-light enhancement. In *British machine vision conference*. British Machine Vision Association.
- Wei, X., Cao, A., Yang, F., & Ma, Z. (2023). Sparse parameterization for epitomic dataset distillation. In A. Oh, T. Naumann, A. Globerson, K. Saenko, M. Hardt, & S. Levine (Eds.), *Advances in neural information processing systems* (pp. 50570–50596). Curran Associates, Inc. (vol. 36).
- Xu, X., Wang, R., Fu, C.-W., & Jia, J. (2022). Snr-aware low-light image enhancement. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 17714–17724).
- Yang, D., Chen, Z., Wang, Y., Wang, S., Li, M., Liu, S., Zhao, X., Huang, S., Dong, Z., Zhai, P. et al. (2023a). Context de-confounded emotion recognition. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 19005–19015).
- Yang, W., Wang, S., Fang, Y., Wang, Y., & Liu, J. (2020). From fidelity to perceptual quality: A semi-supervised approach for low-light image enhancement. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 3063–3072).
- Yang, W., Wang, W., Huang, H., Wang, S., & Liu, J. (2021). Sparse gradient regularized deep retinex network for robust low-light image enhancement. *IEEE Transactions on Image Processing*, 30, 2072–2086.
- Yang, Y., Yuan, H., Li, X., Lin, Z., Torr, P., & Tao, D. (2023b). Neural collapse inspired feature-classifier alignment for few-shot class incremental learning. *arXiv preprint arXiv:2302.03004*.
- Yao, M., Huang, J., Jin, X., Xu, R., Zhou, S., Zhou, M., & Xiong, Z. (2023). Generalized lightness adaptation with channel selective normalization. In *Proceedings of the IEEE/CVF international conference on computer vision* (pp. 10668–10679).
- Yu, H., Huang, J., Zhao, F., Gu, J., Loy, C. C., Meng, D., Li, C. et al. (2022). Deep fourier up-sampling. *Advances in Neural Information Processing Systems*, 35, 22995–23008.
- Yu, J., Zhuge, Y., Zhang, L., Hu, P., Wang, D., Lu, H., & He, Y. (2024). Boosting continual learning of vision-language models via mixture-of-experts adapters. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 23219–23230).
- Zelterman, D. (2001). Causality: Models, reasoning, and inference. *Technometrics: A Journal of Statistics for the Physical, Chemical, and Engineering Sciences*, 43(2), 239–240.
- Zenke, F., Poole, B., & Ganguli, S. (2017). Continual learning through synaptic intelligence. In *Proceedings of the 34th international conference on machine learning - volume 70 ICML'17* (p. 3987–3995). JMLR.org.
- Zhang, H., Cisse, M., Dauphin, Y. N., & Lopez-Paz, D. (2018). Mixup: Beyond empirical risk minimization. In *International conference on learning representations*.
- Zhou, M., Huang, J., Guo, C.-L., & Li, C. (2023). Fourmer: An efficient global modeling paradigm for image restoration. In A. Krause, E. Brunskill, K. Cho, B. Engelhardt, S. Sabato, & J. Scarlett (Eds.), *Proceedings of the 40th international conference on machine learning* (pp. 42589–42601). PMLR (vol. 202). Proceedings of Machine Learning Research.